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Monetary Policy and Fragility in Corporate Bond Mutual Funds

Contents

1 Big picture

- **Question.** How do Federal Funds Target rate changes affect corporate bond mutual fund flows, and when does monetary policy create fragility in these funds?
- **Main fact.** Corporate bond mutual funds experience aggregate outflows around increases in the Federal Funds Target rate, both in monthly data and in narrow windows around FOMC meetings.
- **Core mechanism.** Fund net asset values (NAVs) are stale because corporate bonds trade infrequently. When market participants learn about future rate increases before FOMC meetings, fund shares can become temporarily overpriced relative to intrinsic value. Investors redeem before NAVs fully adjust.
- **Why FOMC windows matter.** Changes in Eurodollar futures before FOMC meetings predict future Federal Funds Target rate changes, future NAV declines, and fund outflows. This links market learning, stale NAVs, and redemptions in a tight event window.
- **Fragility channel.** Redeeming investors benefit from stale overvalued NAVs but impose liquidation costs on remaining investors. This creates strategic complementarities and a first-mover advantage.
- **Cross-sectional prediction.** Outflow sensitivity to monetary-policy news is stronger for high-staleness funds when market conditions are liquid.
- **Counterintuitive result.** Staleness can stabilize outflows in distressed conditions. When liquidity is low, investors are already inclined to redeem because others' redemptions are costly; a stale NAV that adjusts less upward can reduce redemption incentives.
- **Illiquidity result.** Fund-level and market-level illiquidity amplify the outflow response to monetary-policy tightening.
- **Policy-environment result.** The outflow–ΔFFTar sensitivity differs across loose and tight monetary policy regimes. Loose policy environments strengthen the mechanism in liquid settings but weaken it in distressed settings.
- **Economic takeaway.** Monetary policy can systematically exacerbate fragility in corporate bond mutual funds through stale NAVs and strategic redemptions. Reducing staleness is not always stabilizing; its effect depends on liquidity and the policy environment.

2 Data and measurement

2.1 Corporate bond mutual fund data

- Fund-level data come from CRSP and Morningstar.
- The daily sample covers January 2009 to June 2023.
- The monthly sample covers January 1992 to June 2023.
- The main object is corporate bond mutual fund outflows, measured as a percentage of fund total net assets (TNA).
- Fund controls include lagged performance, return, TNA, expense ratio, cash and government bond holdings, high-yield status, and fund fixed effects where appropriate.

Interpretation: The data combine high-frequency event-window flow measures with longer-run monthly flow data. This allows the paper to connect FOMC learning and stale pricing with the broader monetary-policy flow sensitivity.

2.2 Federal Funds Target rate and futures data

- ΔFFTar is the change in the Federal Funds Target rate at or around FOMC meetings.
- The event-window proxy for market expectations is the change in Eurodollar futures before the FOMC meeting.
- Eurodollar futures changes are used because they predict future Federal Funds Target rate changes in the post-2009 sample.

Interpretation: The futures-market variation is central to identification. It captures market learning about future rate increases before the official announcement.

2.3 Outflows and event windows

The paper measures cumulative fund outflows in several windows around each FOMC meeting:

$$\text{Outflow}_{i,(t_1,t_2)} = \sum_{d=t_1}^{t_2} \text{Outflow}_{i,d}. \tag{1}$$

The main windows are:

- $(-5,-1]$: before the FOMC meeting;
- $(-1,5]$: from just before to after the FOMC meeting;
- $(5,15]$: the later post-meeting period.

Interpretation: The mechanism predicts outflows before and shortly after meetings, when the market learns rate news but NAVs have not fully adjusted. Later windows should show weaker effects if mispricing dissipates.

2.4 NAV staleness

The paper proxies NAV staleness by the proportion of trading days before an FOMC meeting on which the fund's NAV does not move:

$$Staleness_i = \frac{\#\{d : \Delta NAV_{i,d} = 0\}}{\#\{\text{trading days before FOMC}\}}. \tag{2}$$

Funds above the sample median are classified as high-stale; funds below the median are low-stale.

Interpretation: Corporate bond funds have much more stale NAVs than equity funds because corporate bonds trade infrequently and daily NAVs may be based on stale transaction prices or vendor estimates.

2.5 Liquidity measures

- **Fund-level liquidity:** cash and government bond holdings. Funds with lower cash/government bond holdings are more illiquid.
- **Market-level liquidity:** VIX-based market conditions. High VIX months proxy for illiquid or distressed corporate bond market conditions.

Interpretation: Illiquidity raises the cost imposed on remaining investors when others redeem. It therefore amplifies strategic redemption incentives.

3 Models and empirical specifications

3.1 Aggregate monthly and annual flow regression

The aggregate regression is

$$Annual\ Outflow_t = \beta \Delta \text{FFTar}_t + \Gamma Controls_t + \varepsilon_t. \tag{3}$$

The same structure is estimated at the monthly frequency.

Interpretation: A positive coefficient means that rate increases are associated with larger fund outflows. The comparison with bank deposits shows that corporate bond fund flows are especially sensitive to rate changes.

3.2 Predicting target-rate changes

The first premise is that market rates reveal information before FOMC meetings:

$$\Delta \text{FFTar}_{[-1,1]} = \beta \Delta Futures_{(r+5,-1]} + \varepsilon_r. \tag{4}$$

The paper estimates this using Federal Funds futures and Eurodollar futures.

Interpretation: If futures changes predict target-rate changes, investors can learn about likely policy tightening before the meeting and act before official NAV adjustments are complete.

3.3 NAV mispricing regression

The NAV-predictability regression is

$$\Delta NAV_{i,(t_1,t_2)} = \beta \Delta Eurodollar_{(r+5,-1]} + \Gamma Controls^F_{i,r-1} + \alpha_i + \varepsilon_{i,r}. \tag{5}$$

Interpretation: If Eurodollar futures changes predict later NAV declines, then fund shares were temporarily overpriced before or around the FOMC meeting. This is the key stale-pricing evidence.

3.4 Strategic redemption regression

The outflow regression uses the same policy-information variable:

$$\text{Outflow}_{i,(t_1,t_2)} = \beta \Delta Eurodollar_{(r+5,-1]} + \Gamma Controls^F_{i,r-1} + \alpha_i + \varepsilon_{i,r}. \tag{6}$$

Interpretation: A positive coefficient shows that investors redeem when policy news implies future NAV declines. The effect should be stronger for high-staleness funds if redemption is driven by stale overpricing.

3.5 Model of redemption with stale NAVs

Investors choose whether to redeem at the posted NAV or stay in the fund. Their payoff depends on:

- the current NAV p_1^{NAV};
- the intrinsic fund value p_1;
- the future NAV adjustment ΔNAV;
- the liquidation cost or illiquidity parameter;
- the mass of other redeeming investors.

In equilibrium, investors redeem when the ratio of stale NAV to intrinsic value is sufficiently high:

$$\frac{p_1^{NAV}}{p_1} \geq \text{threshold}. \tag{7}$$

Interpretation: Redemption is not only arbitrage against mispricing. It is strategic: investors care about other investors' redemptions because asset liquidation imposes costs on those who remain.

3.6 Main hypotheses

1. **Stale-pricing hypothesis.** Monetary-policy news predicts future NAV adjustment and outflows around FOMC meetings.
2. **Illiquidity hypothesis.** Illiquidity strengthens the outflow–ΔFFTar sensitivity by increasing redemption externalities.
3. **Staleness-under-distress hypothesis.** Staleness weakens outflows when liquidity is low but strengthens them when liquidity is high.
4. **Policy-environment hypothesis.** The outflow–ΔFFTar sensitivity is weaker in low-interest-rate environments when liquidity is low, and stronger when liquidity is high.

Interpretation: These hypotheses separate three forces: stale arbitrage, strategic redemption externalities, and the monetary-policy environment.

4 Identification and empirical design

4.1 Why the FOMC event window is useful

- FOMC meetings provide repeated policy-news events.
- Futures markets reveal information about rate changes before the meetings.
- Corporate bond fund NAVs adjust with delay because underlying bond prices are stale.
- Daily fund flows allow the authors to observe strategic redemption before and after meetings.

Interpretation: The design isolates a narrow channel: investors learn about future policy tightening, stale NAVs do not adjust immediately, and redemptions follow.

4.2 Why high- versus low-staleness funds matter

- High-staleness funds should be more exposed to stale mispricing.
- If the mechanism is truly stale NAV arbitrage, outflows should be more sensitive to policy news for high-staleness funds in liquid conditions.
- If staleness plays a stabilizing role in distress, the high-staleness effect should weaken or reverse under illiquid conditions.

Interpretation: The cross-sectional comparison across stale and non-stale funds is the paper's main mechanism test.

4.3 Alternative explanations addressed

- **Aggregate portfolio reallocation:** the paper finds no similar outflow–ΔFFTar pattern in equity funds.
- **Reaching for yield:** robustness tests show that the event-window findings are not driven purely by reaching-for-yield behavior.
- **Ordinary performance-flow relation:** fund fixed effects and lagged performance controls address the standard sensitivity of flows to past returns.

- **Fund liquidity only:** the paper separates staleness from fund-level and market-level liquidity.

Interpretation: The mechanism is not simply that investors dislike bonds when rates rise. It is the interaction of policy news, stale NAVs, and redemption incentives.

5 Tables and figures

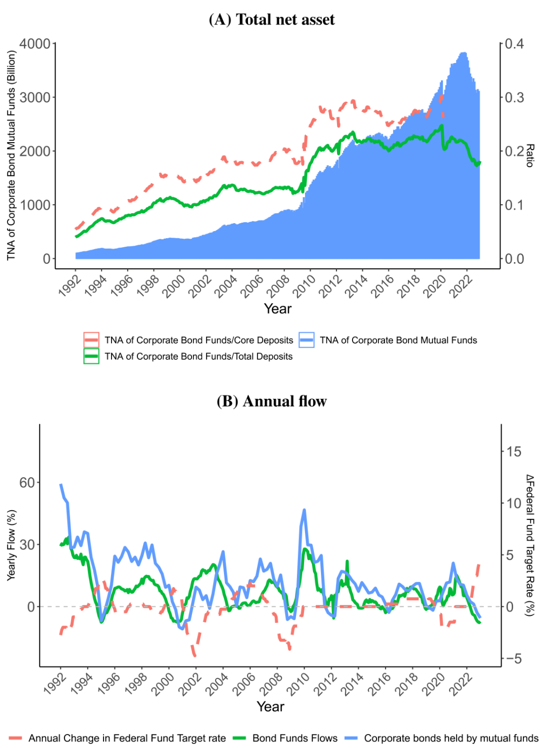
The figures and tables below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and Tables 1–2: aggregate facts and basic model payoffs

Read:

- Figure 1 shows that corporate bond mutual fund assets rose sharply from the 1990s to 2023, reaching several trillion dollars.
- Panel B shows that bond fund flows co-move with Federal Funds Target rate changes, especially during rate-tightening periods.
- Table 1 summarizes the model's payoff comparison between redeeming and staying.
- Table 2 reports summary statistics for daily and monthly fund-level data. Daily outflows average around −0.033% and monthly outflows average around −0.727%.

Economic message: Corporate bond funds have become systemically important non-bank intermediaries, and their demandable shares create a natural setting for monetary-policy-induced fragility.



(a) Figure 1: fund assets and flows.

Table 2

Summary statistics of fund characteristics. The table provides a summary of the characteristics of corporate bond mutual funds, utilizing data from CSRP and Morningstar. The daily data spans from January 2009 to June 2023, covering 3182 funds, while the monthly data encompasses January 1992 to June 2023 and includes 6251 unique fund share classes across 2447 distinct funds. "OutFlows" is the fund outflows in a given month or a given period around FOMC meetings in percentage points. Monthly (Daily) return is the monthly (daily) net fund return in percentage points. The table includes the following characteristics: "TNA" indicates the total net assets of the funds, "Age" represents the number of years since the fund's inception as recorded in the CRSP database, "Expense" denotes the fund's expense ratio, expressed in percentage points, "Cash Holdings" reflects the proportion of fund assets held in cash, presented as a percentage, "Government Bond Holding" represents the proportion of fund assets invested in government bonds, also expressed as a percentage and "Maturity" indicates the weighted average maturity of the fund's investments, measured in years. Perf, η_B, η_M are coefficients from regression (11) for each fund share. Exchange-traded funds and exchange-traded notes have been excluded from the analysis using the CRSP mutual fund database. To address the impact of outliers, all continuous variables have been winsorized at the 1% quantile from each tail.

Panel A: Daily data (January 2009 to June 2023)								
	N	Mean	Std Dev	P5	P25	Median	P75	P95
OutFlows (%)	7,785,862	−0.033	0.422	−0.531	−0.089	−0.003	0.065	0.360
Daily return (%)	8,282,198	0.014	0.246	−0.385	−0.096	0.000	0.107	0.411
$OutFlows_{(-5,-1]}$ (%)	168,019	−0.155	1.558	−2.338	−0.435	−0.026	0.269	1.578
$OutFlows_{(-1,5]}$ (%)	168,047	−0.293	2.319	−3.595	−0.770	−0.098	0.396	2.429
$OutFlows_{(5,15]}$ (%)	166,113	−0.389	2.839	−4.581	−0.976	−0.099	0.542	2.935
Panel B: Monthly data (January 1992 to June 2023)								
	N	Mean	Std Dev	P5	P25	Median	P75	P95
Monthly outflows (%)	745,930	−0.727	8.038	−12.708	−1.690	0.175	1.631	7.610
Monthly return (%)	745,930	0.319	1.461	−2.171	−0.250	0.315	1.013	2.582
TNA (million)	745,930	461.412	1361.585	0.200	8.300	54.100	263.700	2153.955
Age (years)	745,930	10.022	8.667	0.992	3.671	7.740	14.159	25.997
Expense (%)	615,256	0.975	0.491	0.320	0.610	0.860	1.300	1.870
Cash holding (%)	660,539	2.582	11.409	−14.070	0.000	1.880	4.840	18.220
Government bond holding (%)	660,539	13.100	17.443	0.000	0.000	4.060	22.210	49.040
Maturity (years)	490,966	10.044	5.523	2.900	6.300	9.200	13.200	18.400
Perf (%)	672,415	−0.021	0.356	−0.579	−0.143	−0.029	0.083	0.579
η_B	672,415	0.634	0.478	−0.058	0.271	0.678	0.960	1.320
η_M	672,415	0.131	0.164	−0.022	0.012	0.064	0.210	0.486

(b) Table 2: summary statistics.

Table 1
The payoff of investors at T_2.

	$\lambda \le \frac{\mathcal{L}p_1}{s\bar{p}_1+(1-s)p_1}$	$\lambda > \frac{\mathcal{L}p_1}{s\bar{p}_1+(1-s)p_1}$
Redeem	$\frac{\text{NAV}}{p_1}$	$\frac{\mathcal{L}p_1}{p_0\lambda} \times \frac{1}{p_1}$
Stay	$\frac{1}{1-\lambda} \times \left(\frac{1}{p_0} - \frac{\lambda\text{NAV}}{\mathcal{L}p_1}\right) + \frac{\psi}{p_0}$	0

5.2 Tables 3–4: aggregate outflow sensitivity and futures predictability

Read:

- Table 3 shows that annual corporate bond fund outflows are highly sensitive to Federal Funds Target rate increases.
- The annual bond-fund coefficient on ΔFFTar is 4.878, while the comparable deposit coefficients are much smaller.
- The monthly bond-fund coefficient is 1.276, implying a sizable monthly outflow response to rate increases.
- Table 4 shows that both Federal Funds futures and Eurodollar futures predict future target-rate changes, with strong coefficients and high significance.

Economic message: Corporate bond funds are more flow-sensitive to monetary tightening than bank deposits, and futures markets reveal policy news before FOMC meetings.

Table 3
Monetary policy and outflows from corporate bond mutual funds and banks. The table displays time series regressions of aggregate outflows from corporate bond mutual funds and banks in relation to FFTar changes. The data spans from January 1992 to June 2023. The first three columns present annual outflows, while the last three columns show monthly outflows. All values are expressed as a percentage of the size of aggregate corporate bond mutual funds (columns 1 and 4), commercial bank deposits (columns 2 and 5), and core deposits (columns 3 and 6). Core deposits comprise small time, saving, demand, and other checkable deposits (the sample stops in April 2020 due to the discontinuation of saving series SVGCBNS in the FRED database). ΔFFTar indicates the percentage point changes in the FFTar rate and shares the same construction window as the dependent variable. Additional macro control variables include the change in the Baa-Aaa Spread, the change in the spread between the 30-year and 1-year treasury yields, and the logarithmic change in the VIX index. These variables also have the same construction window as the dependent variable. "COVID" is an indicator variable with a value of 1 for the year 2020 in the annual regression and for March 2020 in the monthly regression, and 0 otherwise. Coefficients (standard errors) are reported in shaded (unshaded) rows. Standard errors in brackets are computed with Newey–West standard errors with 12 lags. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

	Annual outflows (%)			Monthly outflows (%)		
	Bond funds (1)	Deposits (2)	Core deposits (3)	Bond funds (4)	Deposits (5)	Core deposits (6)
ΔFFTar	4.878***	0.688	1.918**	1.276***	0.676***	0.712*
	(1.384)	(0.829)	(0.741)	(0.380)	(0.224)	(0.413)
ΔBaa-Aaa spread	9.208***	0.582	−0.066	2.766***	0.564	−1.388***
	(1.566)	(0.993)	(0.903)	(0.492)	(0.516)	(0.371)
Δ30Y-1Y spread	2.055	−0.188	0.877	0.209	−0.002	−0.283
	(1.605)	(0.939)	(0.824)	(0.434)	(0.174)	(0.260)
Δlog(VIX)	−0.481	−0.334	1.460*	−0.179	0.001	0.310*
	(1.348)	(1.022)	(0.779)	(0.334)	(0.223)	(0.182)
COVID	7.686***	−10.106***	−1.082	4.247***	−4.922***	−1.700**
	(2.036)	(2.207)	(1.211)	(0.787)	(0.540)	(0.746)
Constant	−6.891***	−6.234***	−6.370***	−0.507***	−0.508***	−0.534***
	(0.978)	(0.499)	(0.435)	(0.092)	(0.047)	(0.044)
Observations	378	378	339	378	378	339
Adjusted R^2	0.439	0.332	0.272	0.126	0.131	0.156

(a) Table 3: aggregate outflows and policy changes.

Table 4
Predictive regressions for federal fund target rate changes. The table presents the results of predictive regressions to predict FFTar changes within a window of [−1, 1] around FOMC meetings, using market data prior to the meetings. The predictor variable is changes in rates for the Federal Funds Futures (columns 1 and 3) and Eurodollar Futures (columns 2 and 4) within the (τ+5, −1] window. Here, τ represents the date of the preceding FOMC meeting. The analysis employs two different sample windows: January 1992 to June 2023 and January 2009 to June 2023. Coefficients (standard errors) are reported in shaded (unshaded) rows. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

	$\Delta FFTar_{[-1,1]}$			
	Year ≥ 1992		Year ≥ 2009	
	(1)	(2)	(3)	(4)
$\Delta \text{FFuture}_{(\tau+5,-1]}$	0.643***		0.501***	
	(0.056)		(0.065)	
$\Delta \text{EuroDollar}_{(\tau+5,-1]}$		0.687***		0.886***
		(0.053)		(0.076)
Constant	0.003	−0.003	0.018	0.014
	(0.011)	(0.010)	(0.015)	(0.012)
Observations	281	281	125	125
Adjusted R^2	0.320	0.372	0.323	0.521

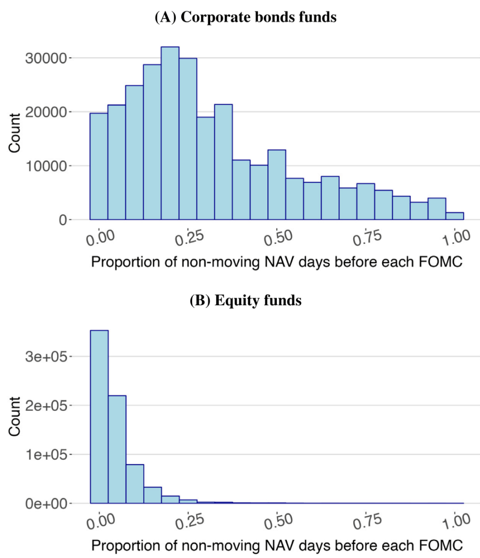
(b) Table 4: futures predict target-rate changes.

5.3 Figure 2 and Table 5: staleness and NAV mispricing

Read:

- Figure 2 shows that corporate bond funds have far more non-moving NAV days than equity funds before FOMC meetings.
- Table 5 shows that Eurodollar futures changes predict future NAV declines for high-staleness funds.
- The predictive power persists until five days after the FOMC meeting for high-staleness funds.
- Low-staleness funds show weaker or insignificant NAV predictability.

Economic message: NAVs of high-staleness corporate bond funds do not fully incorporate policy news promptly. This creates a temporary stale-pricing wedge that investors can exploit.



(a) Figure 2: staleness distributions.

Table 5

NAV changes around FOMC meetings. This table compares the responsiveness of fund NAVs to market information on monetary policy changes around FOMC meetings, separately for high-staleness funds to low-staleness funds (which we refer to as High-stale and Low-stale to conserve space). The dependent variables are the logarithmic changes in NAV for each fund share i within four different time windows around FOMC meetings: (τ+5,−5], (−5,−1], (−1,5], and (5,15], with 0 representing the date of the FOMC meeting, and τ indicating the date of the previous FOMC meeting. For each time window, the information revealed by the Eurodollar Future rates are measured within the respective windows. Funds that exhibit a higher (lower) proportion of non-moving NAV days in the non-FOMC window leading up to the preceding FOMC meeting, compared to the median, are classified as high-(low-)staleness funds. We include one-year lagged fund characteristics, such as the total net asset in log scale, expense ratios, percentage of cash and government bond holdings, outflows from the past five days, and high-yield fund indicator, as controls, denoted as $\text{Controls}^F_{i,t-1}$. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. Each observation is weighted by the previous year's end-of-year fund TNA. Standard errors are clustered at each FOMC meeting and the fund share level. Coefficients (standard errors) are reported in shaded (unshaded) rows. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

	$\Delta NAV_{i,(\tau+5,-5]}$		$\Delta NAV_{i,(-5,-1]}$		$\Delta NAV_{i,(-1,5]}$		$\Delta NAV_{i,(5,15]}$	
	High-stale (1)	Low-stale (2)	High-stale (3)	Low-stale (4)	High-stale (5)	Low-stale (6)	High-stale (7)	Low-stale (8)
$\Delta Eurodollar_{(\tau+5,-5]}$	−1.683***	−3.744**	−0.687**	−0.912				
	(0.591)	(1.490)	(0.296)	(0.559)				
$\Delta Eurodollar_{(\tau+5,-1]}$					−0.741***	−0.732		
					(0.213)	(0.508)		
$\Delta Eurodollar_{(\tau+5,5]}$							0.137	−0.467
							(0.290)	(0.365)
$\text{Controls}^F_{i,t-1}$	✓	✓	✓	✓	✓	✓	✓	✓
Fund FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	80,229	98,669	80,227	98,665	83,144	102,380	83,151	102,431
Adjusted R^2	0.140	0.189	0.090	0.067	0.096	0.021	0.085	0.078

(b) Table 5: NAV predictability around FOMC meetings.

5.4 Table 6: monetary-policy-induced redemptions

Read:

- Table 6 shows that Eurodollar futures changes predict fund outflows around FOMC meetings.
- The effect is especially strong for high-staleness funds in the windows $(-5, -1]$ and $(-1, 5]$.
- The outflow response is weaker after the mispricing window dissipates.
- The paper interprets the magnitudes as economically large: a 25-basis-point predicted rate increase is associated with meaningful event-window outflows.

Economic message: Investors redeem when policy news predicts stale NAV declines. This is direct evidence of strategic redemption from temporarily overpriced fund shares.

Table 6

Monetary policy-induced fragility. This table examines how fund flows respond to market information regarding monetary policy changes around FOMC meetings, specifically comparing the responses of high-staleness funds to low-staleness funds (which we refer to as High-stale and Low-stale to conserve space). The dependent variables are the cumulative fund outflows, measured in percentage points, for each fund share i within three different time windows around FOMC meetings: (−5,−1], (−1,5], and (5,15], where 0 represents the date of the FOMC meeting. For each time window, the information revealed by the Eurodollar Future rates is measured within the respective windows (τ+5,−5], (τ+5,−1], and (τ+5,5], where τ indicates the date of the previous FOMC meeting. Funds that exhibit a higher (lower) proportion of non-moving NAV days in the non-FOMC window leading up to the preceding FOMC meeting, compared to the median, are classified as high-(low-)staleness funds. The interaction terms measure the difference in the outflow-rate relationship between high-staleness and low-staleness funds. We include one-year lagged fund characteristics, such as the total net asset in log scale, expense ratios, percentage of cash and government bond holdings, outflows from the past five days, and high-yield fund indicator as controls, denoted as $\text{Controls}^F_{i,t-1}$. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. Each observation is weighted by the previous year's end-of-year fund TNA. Standard errors are clustered at each FOMC meeting and the fund share level. Coefficients (standard errors) are reported in shaded (unshaded) rows. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

	$\text{OutFlows}_{i,(-5,-1]}$			$\text{OutFlows}_{i,(-1,5]}$			$\text{OutFlows}_{i,(5,15]}$		
	High-stale (1)	Low-stale (2)	All (3)	High-stale (4)	Low-stale (5)	All (6)	High-stale (7)	Low-stale (8)	All (9)
$\Delta\text{Eurodollar}_{(\tau+5,-5]}$	0.862***	0.277***	0.277***						
	(0.135)	(0.097)	(0.097)						
$\Delta\text{Eurodollar}_{(\tau+5,-5]} \times \mathbb{1}(\text{High-stale})$			0.585***						
			(0.142)						
$\Delta\text{Eurodollar}_{(\tau+5,-1]}$				1.018***	0.515***	0.515***			
				(0.359)	(0.133)	(0.133)			
$\Delta\text{Eurodollar}_{(\tau+5,-1]} \times \mathbb{1}(\text{High-stale})$						0.503*			
						(0.289)			
$\Delta\text{Eurodollar}_{(\tau+5,5]}$							0.766***	0.435***	0.435***
							(0.211)	(0.118)	(0.118)
$\Delta\text{Eurodollar}_{(\tau+5,5]} \times \mathbb{1}(\text{High-stale})$									0.331
									(0.226)
$\text{Controls}^F_{i,t-1}$	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fund FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	80,262	98,718	178,980	83,133	102,439	185,572	82,632	101,936	184,568
Adjusted R^2	0.217	0.266	0.246	0.262	0.287	0.277	0.231	0.208	0.217

outflows driven by mispricing dissipate within 5 days after the FOMC ... estimate the mechanism in the subsequent five days. In our sample

5.5 Table 7 and Table 8: asymmetric response and illiquidity amplification

Read:

- Table 7 shows that the monthly outflow response is stronger in months with non-negative Federal Funds Target rate changes.
- This asymmetry confirms that the channel is about tightening news rather than symmetric policy changes.
- Table 8 shows that illiquid funds and illiquid market conditions exhibit stronger outflow sensitivity.
- Fund-level liquidity and market-level liquidity both matter.

Economic message: Redemption externalities are stronger when liquidation is costly. Monetary tightening therefore creates larger fragility precisely when liquidity conditions are weak.

Table 7

Asymmetric flow response and monetary policy-induced fragility (monthly evidence). This table examines the impact of monetary policy changes on fund flows of corporate bond mutual funds from January 1992 to June 2023. Only months with FOMC meetings are included in the analysis. The table studies the outflow-rate relationship in the entire sample (columns 1-2), in months with non-negative FFTar moves (columns 3-4), and in months with non-positive FFTar moves (columns 5-6). $\text{OutFlow}_{i,m}$ represents the outflows of fund share i in month m, and ΔFFTar_m denotes the percentage point changes in FFTar. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. The macro controls, $\Delta\text{Controls}^M_m$, include the change in the Baa-Aaa Spread, the change in the spread between the 30-year and 1-year treasury yields, and the logarithmic change in the VIX index. Fund characteristics encompass the previous month's performance, return, TNA on a logarithmic scale, expense ratio, percentage of cash and government bond holdings, and a high-yield fund indicator. Each observation is weighted by the TNA value of the fund from the previous month. Coefficients (standard errors) are reported in shaded (unshaded) rows. Standard errors are clustered at the fund share and month levels. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

	$\text{OutFlow}_{i,m}(\%)$ in Months with FOMC meetings					
	All		$\Delta\text{FFTar}_m \geq 0$		$\Delta\text{FFTar}_m \leq 0$	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔFFTar_m	1.299***	0.742***	2.114***	1.511***	0.552*	0.212
	(0.258)	(0.246)	(0.305)	(0.421)	(0.311)	(0.377)
$\Delta\text{Controls}^M_m$	✓	✓	✓	✓	✓	✓
$\text{Controls}^F_{i,t-1}$		✓		✓		✓
Fund FE		✓		✓		✓
Observations	511,253	336,848	458,772	310,880	410,964	279,079
Adjusted R^2	0.022	0.090	0.014	0.081	0.019	0.092

(a) Table 7: asymmetric response to tightening.

Table 8

The amplifying effect of illiquidity on monetary policy-induced fragility. The table investigates the impact of fund illiquidity on monetary policy-induced fragility around FOMC meetings, with separate analyses conducted for daily and monthly data presented in Panel A and Panel B, respectively. In Panel A, for each FOMC meeting, we classify funds whose last year's percentage holding of liquid assets (cash and government bonds) is higher-(lower-)than-sample median as liquid (illiquid) funds. The dependent variables are the cumulative fund outflows, measured in percentage points, for each fund share i within three different time windows around FOMC meetings: (−5,−1], (−1,5], and (5,15], with 0 representing the date of the FOMC meeting. The predicted changes in FFTar, also in percentage points, are based on $\Delta\text{Eurodollar}(\tau+5,-5]$, denoted as $\widehat{\Delta\text{FFTar}}_{[-1,1]}$. We include one-year lagged fund characteristics, such as the total net asset in log scale, expense ratios, high-staleness fund indicator, high-yield fund indicator as controls, denoted as $\text{Controls}^F_{i,t-1}$. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. In Panel B, the analysis focuses on months with FOMC meetings and non-negative Federal Fund Target rate moves. High (Low) VIX months refer to months with a VIX index above (below) the top (bottom) tercile of the sample. Low (High) CashBond funds denote funds with a proportion of cash and government bond holdings below (above) the bottom (top) tercile within each Lipper objective category of each year. Fund characteristics encompass the previous month's performance, return, TNA on a logarithmic scale, expense ratio, a high-staleness fund indicator, and a high-yield fund indicator. $\Delta\text{Controls}^M_m$ is the same as in Table 7. Each observation is weighted by the TNA value of the fund from the previous month. Coefficients (standard errors) are reported in shaded (unshaded) rows. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

Panel A: Daily evidence

	$\text{OutFlows}_{i,(-5,-1]}$			$\text{OutFlows}_{i,(-1,5]}$			$\text{OutFlows}_{i,(5,15]}$		
	Illiquid (1)	Liquid (2)	All (3)	Illiquid (4)	Liquid (5)	All (6)	Illiquid (7)	Liquid (8)	All (9)
$\widehat{\Delta\text{FFTar}}_{[-1,1]}$	0.873***	0.533***	0.533***	1.699***	1.073***	1.073***	1.905***	1.142***	1.142***
	(0.206)	(0.109)	(0.110)	(0.311)	(0.279)	(0.279)	(0.396)	(0.314)	(0.314)
$\widehat{\Delta\text{FFTar}}_{[-1,1]} \times \mathbb{1}(\text{Illiquid funds})$			0.340			0.626**			0.763***
			(0.218)			(0.290)			(0.240)
$\text{Controls}^F_{i,t-1}$	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fund FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	90,506	88,549	179,055	90,515	88,587	179,102	90,056	88,097	178,153
Adjusted R^2	0.115	0.086	0.099	0.136	0.095	0.112	0.146	0.107	0.124

Panel B: Monthly evidence

	$\text{OutFlow}_{i,m}(\%)$ in Months with FOMC meetings & $\Delta\text{FFTar}_m \geq 0$					
	Liquid vs. illiquid market condition			Liquid vs. illiquid funds		
	High VIX (1)	Low VIX (2)	All (3)	Low CashBond (4)	High CashBond (5)	All (6)
ΔFFTar_m	2.604***	−0.698	−0.698	2.458***	0.149	0.149
	(0.737)	(0.746)	(0.743)	(0.582)	(1.085)	(1.085)
$\Delta\text{FFTar}_m \times \mathbb{1}(\text{High VIX})$			3.302***			
			(1.087)			
$\Delta\text{FFTar}_m \times \mathbb{1}(\text{Low CashBond})$						2.309**
						(1.093)
$\Delta\text{Controls}^M_m$	✓	✓	✓	✓	✓	✓
$\text{Controls}^F_{i,t-1}$	✓	✓	✓	✓	✓	✓
Fund FE	✓	✓	✓	✓	✓	✓
Observations	78,864	86,897	165,761	77,515	73,981	151,496
Adjusted R^2	0.128	0.156	0.141	0.145	0.092	0.114

(b) Table 8: illiquidity amplifies fragility.

5.6 Table 9 and Table 10: staleness under distress and policy environment

Read:

- Table 9 shows that staleness mitigates outflows when liquidity is low or market conditions are distressed.
- The negative triple-interaction terms indicate that high-staleness funds experience a weaker illiquidity-amplified outflow response.
- Table 10 shows that the monetary-policy environment matters: loose policy and tight policy regimes interact differently with liquidity.
- The results suggest that policy effects depend jointly on staleness, liquidity, and the level of interest rates.

Economic message: Staleness is not always destabilizing. In distressed markets, stale NAVs can dampen redemption incentives, which complicates policy prescriptions aimed at eliminating NAV staleness.

Table 9

The stabilizing effect of staleness on monetary policy-induced fragility in distress. The table examines the impact of fund staleness on monetary policy-induced fragility in illiquid funds or during illiquid periods, using daily and monthly data presented in Panel A and B, respectively. In Panel A, the focus is on funds with a last year's percentage holding of liquid assets (cash and government bonds) below the sample median, which are categorized as illiquid funds. Funds experiencing a higher (lower) proportion of non-moving NAV days in the non-FOMC window leading up to the preceding FOMC meeting, compared to the median, are classified as high-staleness (low-staleness) funds (which we refer to as high-stale and low stale funds to conserve space). The dependent variables are the cumulative fund outflows, measured in percentage points, for each fund share i within three different time windows around FOMC meetings: (−5,−1], (−1,5], and (5,15], with 0 representing the date of the FOMC meeting. The predicted changes in FFTar, also in percentage points, are based on ΔEurodollar$(\tau+5,-5]$, denoted as $\widehat{\Delta \text{FFTar}_{[-1,1]}}$. We include one-year lagged fund characteristics, such as the total net asset in log scale, expense ratios, and high-yield fund indicator as controls, denoted as $\text{Controls}^F_{i,t-1}$. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. In Panel B, the analysis is conducted on months with FOMC meetings and non-negative Federal Fund Target rate moves. The sub-sample of illiquid months includes months with a VIX index above the top tercile of the sample. Similarly, the sub-sample of illiquid funds comprises funds with a proportion of cash and government bond holdings below the bottom tercile within each Lipper objective category of each year. High-staleness (low-staleness) funds are identified as those with a proportion of non-moving NAV days in the previous month higher (lower) than the top (bottom) tercile. Fund characteristics encompass the previous month's performance, return, TNA on a logarithmic scale, expense ratio, and a high-yield fund indicator. $\Delta\text{Controls}^M_m$ is the same as in Table 7. Each observation is weighted by the TNA value of the fund from the previous month. Notably, two-way interactions are not reported in the table. Coefficients (standard errors) are reported in shaded (unshaded) rows. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

Panel A: Daily evidence

	$\text{OutFlows}_{i,(-5,-1]}$	$\text{OutFlows}_{i,(-1,5]}$	$\text{OutFlows}_{i,(5,15]}$
	(1)	(2)	(3)
$\widehat{\Delta \text{FFTar}_{[-1,1]}}$	0.390***	0.845***	0.853***
	(0.097)	(0.260)	(0.315)
$\widehat{\Delta \text{FFTar}_{[-1,1]}} \times \mathbb{1}$(Illiquid funds)	0.367	0.735***	1.189***
	(0.253)	(0.270)	(0.339)
$\widehat{\Delta \text{FFTar}_{[-1,1]}} \times \mathbb{1}$(Illiquid funds) $\times \mathbb{1}$(High-stale)	−0.328	−0.648*	−0.665
	(0.265)	(0.379)	(0.511)
$\text{Controls}^F_{i,t-1}$	✓	✓	✓
Fund FE	✓	✓	✓
Observations	179,055	179,102	178,153
Adjusted R^2	0.091	0.102	0.111

Panel B: Monthly evidence

	$\text{OutFlow}_{i,m}(\%)$ in Months with FOMC meetings & $\Delta\text{FFTar}_m \geq 0$	
	(1)	(2)
ΔFFTar_m	−1.290	−1.367
	(0.923)	(1.375)
$\Delta\text{FFTar}_m \times \mathbb{1}$(High VIX)	4.122***	
	(1.434)	
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low CashBond)		2.851**
		(1.307)
$\Delta\text{FFTar}_m \times \mathbb{1}$(High VIX) $\times \mathbb{1}$(High-stale)	−2.915*	
	(1.545)	
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low CashBond) $\times \mathbb{1}$(High-stale)		−3.352**
		(1.624)
$\Delta\text{Controls}^M_m$	✓	✓
$\text{Controls}^F_{i,t-1}$	✓	✓
Fund FE	✓	✓
Observations	165,761	151,493
Adjusted R^2	0.141	0.119

(a) Table 9: staleness under distress.

Table 10

The effect of monetary policy environment on monetary policy-induced fragility (monthly evidence). The table examines the impact of the monetary policy environment on monetary policy-induced fragility for corporate bond mutual funds from January 1992 to June 2023. A loose monetary policy environment refers to months with FFTar below the bottom tercile of the sample (Low FFTar), while a tight monetary policy environment refers to months with FFTar above the top tercile of the sample. Panel A compares the effects in liquid versus illiquid market conditions. Illiquid market conditions refer to months with a VIX index above the top tercile of the sample (High VIX months), while the opposite applies to Low VIX months. Panel B compares the effects in liquid versus illiquid funds. We refer to illiquid funds as those holding a proportion of cash and government bond holdings below the bottom tercile within each Lipper objective category of each year (Low CashBond Funds), while the opposite applies to Low CashBond funds. Fund characteristics encompass the previous month's performance, return, TNA on a logarithmic scale, expense ratio, percentage of cash and government bond holdings and a high-yield fund indicator. $\Delta\text{Controls}^M_m$ is the same as in Table 7. Each observation is weighted by the TNA value of the fund from the previous month. To account for the COVID-19 pandemic, we include an indicator variable that is equal to one for the FOMC meeting on March 3, 2020, and zero otherwise. Coefficients (standard errors) are reported in shaded (unshaded) rows. Standard errors are clustered at the fund share and month levels. *, **, *** represent statistical significance at 10%, 5% and 1% level, respectively.

Panel A: Liquid vs. illiquid market condition

	$\text{OutFlow}_{i,m}(\%)$ in Months with FOMC meetings		
	High VIX months (1)	Low VIX months (2)	All sample (3)
ΔFFTar_m	1.708***	−0.275	−0.216
	(0.417)	(0.405)	(0.429)
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low FFTar)	−2.232**	2.508**	−0.065
	(0.960)	(1.221)	(0.987)
$\Delta\text{FFTar}_m \times \mathbb{1}$(High VIX)			1.524***
			(0.469)
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low FFTar) $\times \mathbb{1}$(High VIX)			−2.934**
			(1.206)
$\Delta\text{Controls}^M_m$	✓	✓	✓
$\text{Controls}^F_{i,t-1}$	✓	✓	✓
Fund FE	✓	✓	✓
Observations	70,325	72,116	142,441
Adjusted R^2	0.220	0.190	0.163

Panel B: Liquid vs. illiquid funds

	$\text{OutFlow}_{i,m}(\%)$ in Months with FOMC meetings		
	Low CashBond funds (1)	High CashBond funds (2)	All sample (3)
ΔFFTar_m	1.102*	0.187	0.243
	(0.593)	(0.296)	(0.303)
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low FFTar)	−1.736**	0.970	1.427
	(0.749)	(0.838)	(0.881)
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low CashBond)			0.873
			(0.652)
$\Delta\text{FFTar}_m \times \mathbb{1}$(Low FFTar) $\times \mathbb{1}$(Low CashBond)			−2.807**
			(1.229)
$\Delta\text{Controls}^M_m$	✓	✓	✓
$\text{Controls}^F_{i,t-1}$	✓	✓	✓
Fund FE	✓	✓	✓
Observations	76,936	74,262	151,198
Adjusted R^2	0.189	0.163	0.159

(b) Table 10: monetary policy environment.

6 Short summary

- Corporate bond mutual funds experience sizeable outflows when the Federal Funds Target rate rises.
- Futures markets reveal information about future target-rate increases before FOMC meetings.
- Corporate bond fund NAVs are stale and do not fully adjust to policy news immediately.
- Eurodollar futures changes predict future NAV declines and event-window outflows.
- High-staleness funds have stronger outflow sensitivity in liquid conditions.
- Illiquidity amplifies outflow sensitivity because redeeming investors impose liquidation costs on staying investors.
- Surprisingly, staleness can stabilize outflows during distress by muting NAV movements that would otherwise trigger strategic redemptions.

- Loose versus tight monetary policy environments change the strength and sign of the fragility channel across liquidity states.
- The paper's policy message is nuanced: reducing NAV staleness can backfire during market stress.

7 Conclusion

7.1 Core conclusion

Kuong, O'Donovan, and Zhang show that monetary policy can induce fragility in corporate bond mutual funds through stale NAVs. When investors learn about future rate increases before official target-rate changes, corporate bond fund NAVs adjust slowly, fund shares become temporarily overpriced, and investors strategically redeem.

7.2 Economic intuition

The mechanism combines stale pricing with redemption externalities. In liquid markets, stale overpricing makes investors behave like arbitrageurs and redeem from overpriced fund shares. In illiquid markets, however, investors fear others' redemptions and liquidation costs. In that environment, staleness can dampen the movement in NAVs and sometimes reduce redemption incentives.

7.3 Takeaway

The paper adds a new monetary-policy transmission channel through non-bank financial intermediaries. Corporate bond mutual funds are fragile not only because they hold illiquid assets and offer redeemable shares, but also because their NAVs can be stale exactly when policy news arrives. Policy responses should account for the interaction among staleness, market liquidity, and the monetary-policy environment.